- Waves. If the surface of a pond is disturbed, it sends out waves. Someone watching at a particular location will see the surface oscillate as the wave passes by. This motion can be described as simple harmonic motion. The same goes for sound waves.

- Electromagnetic waves. Just like any other wave, a light wave or a radio wave oscillates when it passes you. The same mathematics that describes the oscillating particle also applies to electromagnetic waves.

The list goes on and on but the math is always the same. Just to have an example in mind, let's picture the oscillator as a weight hanging from a spring. Needless to say, we hardly need quantum mechanics to describe an ordinary weight and spring, so let's imagine a very tiny version of this same system and then quantize it.

10.1 THE CLASSICAL DESCRIPTION

Let's use *y* to denote the height of the hanging weight. We'll choose the origin so that the weight is at $y = 0$ when it's in equilibrium—that is when the weight is hanging at rest. To study this system classically, we can use the Lagrangian method that we learned about in *Volume I*. The kinetic and potential energies are $\frac{1}{2}m\dot{y}^2$ and $\frac{1}{2}ky^2$ respectively.

As you recall, the Lagrangian is the kinetic energy *minus* the potential energy:

$$L = \frac{1}{2}m\dot{y}^2 - \frac{1}{2}ky^2.$$

First, we'll put the Lagrangian into a certain standard form by changing from *y* to another variable that we will call *x*. This coordinate is not something new. It still represents the displacement of the mass. By switching from *y* to *x*, we're just making a convenient change of units. Let's define the new variable as